UNIVERSITI TUNKU ABDUL RAHMAN

ACADEMIC YEAR 2024/2025

SEPTEMBER 2024 EXAMINATION

UECS3253 / UECS3463 WIRELESS APPLICATION DEVELOPMENT

MONDAY, 7 OCTOBER 2024 TIME : 9.00 AM – 11.00 AM (2 HOURS)

BACHELOR OF SCIENCE (HONOURS) SOFTWARE ENGINEERING
BACHELOR OF ENGINEERING (HONOURS) ELECTRONICS
(COMPUTER NETWORKING)

Instructions to Candidates:

This question paper consists of **TWO (2)** questions in Section A and **TWO (2)** questions in Section B.

Section A: Answer **ALL** questions.

Section B: Answer **ONLY ONE (1)** question.

Each question in **Section A** carries 30 marks, question in **Section B** carries 40 marks.

Remark: Should a candidate answer **TWO (2)** questions in **Section B**, marks will only be awarded for the **FIRST** question in the order the candidate submits the answers.

Answer the questions in the answer booklet provided.

UECS3253 / UECS3463 WIRELESS APPLICATION DEVELOPMENT

Section A: Answer **ALL** questions.

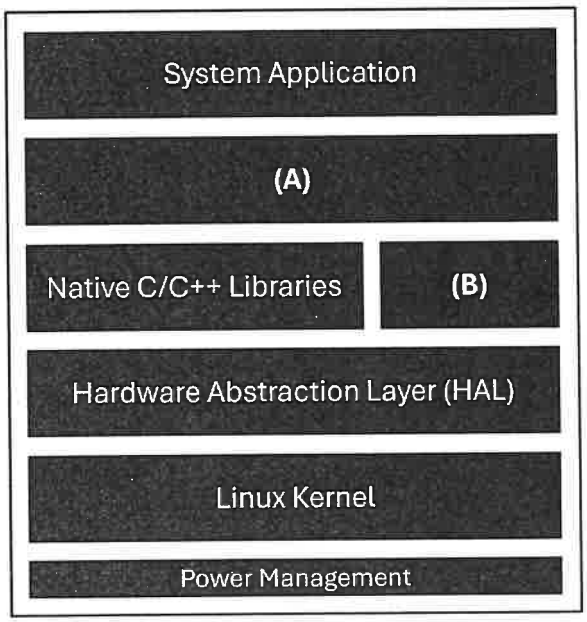
Q1. (a) There are three types of mobile applications.

(i) Suppose you are developing an application. Consider the application is required to have access to the device hardware (e.g., camera and GPS) and works offline, identify **ONE (1)** suitable type of the mobile application. (2 marks)

(ii) Suppose you are developing an application. Consider the application is accessible through the web browser in the mobile device and does not require the user to install new application, identify **ONE (1)** suitable type of the mobile application. (2 marks)

(b) Android is a mobile operating system that is built upon Linux kernel and open-source software.

(i) The figure below shows the Android architecture. By referring to the figure below, identify (A) and (B). (4 marks)



(ii) State **TWO (2)** frameworks/languages that can be used for Android development. (2 marks)

(iii) State **TWO (2)** features of android. (2 marks)

Section A Q1. (Continued)

(c) React Native is a JavaScript-based framework that can be used to create Android applications.

(i) State **TWO (2)** ways to create a component in React Native. (4 marks)

(ii) Functional components can use React Hooks to achieve different React features. State **TWO (2)** React Hooks and describe the effect for each React Hook given. (8 marks)

(iii) By referring to the code segment below, identify the functional component-based implementation for line 4, lines 8-10 and line 15.

```
1:  // Assume the necessary Components were
2:  imported.
3:
4:  export default class App extends Component
5:  {
6:    constructor(props: any) {
7:        super(props);
8:        this.state = {
9:            studentId: 0
10:       };
11:   }
12:   render(){
13:       return(
14:         <CustomComponent
15:           displayId={this.state.studentId}
16:         />
17:       );
18:   }
19: }
```

(6 marks)
[Total : 30 marks]

Section A (Continued)

Q2. (a) SQLite can be used to develop application that incorporate persistent data storage.

(i) Consider the `react-native-sqlite-storage` is not installed, construct the command to install it. (2 marks)

(ii) Explain **THREE (3)** features of `react-native-sqlite-storage`. (6 marks)

(iii) By referring to the code segment below, describe the effect of the code in the application. (4 marks)

```
const methodA = async (
    db: SQLiteDatabase,
    studentId: number
) => {
    const query = ` DELETE FROM studentList
        WHERE studentId = ?`

    try {
        await db.executeSql(query,
[studentId]);
      } catch (error) {
        ToastAndroid.show("Failed");
        throw Error("Failed")
      }
}
```

(iv) Using the code segment below, construct the code to update the record in the database table `studentList` from Table 2.1 to Table 2.2. Display an Alert message when there is an error occurs.

Table 2.1: Original studentList

studentId	studentName	studentAge	programme
1	Student A	20	SE
2	Student B	21	SE

Table 2.2: Updated studentList

studentId	studentName	studentAge	programme
1	Student A	20	SE
2	Student B	22	MH

Section A Q2 (a) (iv). (Continued)

```
const updateStudent = async (
    db: SQLiteDatabase,
    studentId: number,
    studentName: string,
    studentAge: number,
    programme: string
) => {
    /*
      Complete the code here
    */
}
```

(4 marks)

(b) AsyncStorage refers to the data storage method that stores the data in the form of key-value pairs.

(i) Describe **ONE (1)** disadvantage of AsyncStorage. (2 marks)

(ii) By referring to the code segment below, construct the code to store the data in the application using AsyncStorage. Display a console message when there is an error occurs.

Table 2.1: Data to be stored in the device

studentName	studentAge	studyYear
Student C	20	2

```
const storeData = async() => {
    /*
      Complete the code here
    */
}
```

(4 marks)

(c) In React Native, the interface of Geolocation API represents an object that can be used to obtain the position of the device.

(i) By referring to the code segment below, describe the effect of the code in the application.

Section A Q2 (c) (i) (Continued)

```
const methodB = async() => {
    try {
        const granted =
await PermissionsAndroid.request(
            PermissionsAndroid.
PERMISSIONS.ACCESS_FINE_LOCATION,
            {
                'title': 'Title A',
                'message': 'Message',
            }
        )

        if (granted ===
 PermissionsAndroid.RESULTS.GRANTED) {
            console.log('xxx')
        }
        else {
            console.log('yyy')
        }

        return granted
    }
    catch (err) {
        console.warn(err)
    }
}
```

(4 marks)

(ii) By referring to the code segment below, construct the code to retrieve the current position of the device where the application is required to receive a more accurate position, timeout is 5000 milliseconds and maximumAge is 1500 milliseconds.

```
const methodC = () => {
    /*
      Complete the code here
    */
}
```

(4 marks)
[Total : 30 marks]

Section B: Answer **ONLY ONE (1)** question.

Q3. As a software developer, you are developing a mobile application for a HR department that could provide several functionalities, including Employee Update and Employee Retrieve. Particularly, web service will be implemented where a Flask server will be used to process the information, and the React Native application will connect to the server.

(a) In this scenario, web service is implemented. Answer the following questions related to web service.

(i) Other than the Simple Object Access Protocol (SOAP), identify **TWO (2)** communication methods that can be used for web services and then explain each of the communication method. (8 marks)

(ii) Identify **TWO (2)** similarities between Application Programming Interface (API) and web services. (4 marks)

(iii) Identify **THREE (3)** HTTP status code that can be responded by the server. (3 marks)

(b) One of the functionalities is the **Employee Update**. To do so, the application should send the update information to the server using the HTTP PUT method through the URL `https://hrapp.com/api/updateemployee`. The data being sent is a JSON data object containing the employee's ID and updated salary, as shown in the **Table 3.1**.
Remark: Fetch API is used for allowing the React Native application to make HTTP requests to the server.

Table 3.1: Updated Employee Information

Field	Data	Type	Description
employeeId	1234	String	The ID of the employee
salary	3000	String	The salary of the employee

After the information is successfully submitted to the server, the server will process the information and respond to the React Native application with the message as shown below.

Table 3.2: Response from the server

Field	Type	Description
updateID	String	The unique ID assigned to the update process
success	Boolean	A flag indicating the success of the updating

Using the code segment below, construct a code segment that sends the update request and data to the server using the JSON format stated in **Table 3.1**.

Section B Q3 (b) (Continued)

Upon receiving the data, the server starts to update the employee information and send the response to the React Native application to indicate whether or not the update is successful. If the update is successful, the React Native application will display a Toast message and navigate to a screen named "EmployeeDetails" where the updateID will be passed to a route parameter. If the update is unsuccessful, the React Native application will display an Alert message. In the event of error occurs during the communication with the server, the React Native will display an appropriate error message.

```
const _updateEmployee = () => {
  let url = 'https://hrapp.com/api/updateemployee'
  /*
    Complete the code here
  */}
```

(15 marks)

(c) One of the functionalities is the **Employee Retrieve**. To do so, the application should send the request to the server using the HTTP GET method through the URL `https://hrapp.com/api/retrieveemployee`.
Remark: Fetch API is used for allowing the React Native application to make HTTP requests to the server.

After the information is successfully submitted to the server, the server will process the request and respond to the React Native application with the message as shown below.

Table 3.3: Response from the server

Field	Type	Description
employee	Object	An object that consists of the employee information
success	Boolean	A flag indicating the success of the retrieval

Using code segment below, construct the code that sends the retrieve request to the server. Upon receiving the request from the application, the server starts to retrieve the employee information from the database and send the response to the React Native application. If the retrieval is successful, the React Native application will display a Toast message and set the value to the state variable `employee`. If the retrieval is unsuccessful, the React Native application will display an Alert message and navigate to the previous screen. In the event of error occurs during the communication with the server, the React Native will display an appropriate error message.

Section B Q3 (c) (Continued)

```
const [employee, setEmployee] = useState({});

const _retrieveEmployee = () => {
  let      url      =      'https://hrapp.com/api/
retrieveemployee'
  /*
    Complete the code here
  */}
```

(10 marks)
[Total : 40 marks]

Q4. As a software developer, you are developing a mobile application for a food delivery company. The application should provide several functionalities, including real-time driver location tracking and in-app live chat. Particularly, WebSocket will be implemented where a Flask server will be used to process the information, and the React Native application will connect to the server.

(a) In this scenario, WebSocket is implemented. Answer the following questions related to WebSocket.

(i) Explain WebSocket and analyse the reason why the WebSocket is implemented in this scenario in relation to the functionalities provided by the application. (4 marks)

(ii) In addition to the functionalities stated in the Q4, identify **THREE (3)** sample use cases for WebSocket. (6 marks)

(iii) Consider the `socket.io-client` is required for the project, construct the command to install `socket.io-client` to the React Native project folder. (2 marks)

(b) In React Native, `socket.io-client` module enables the developer to turn the React Native applications into a client application that connects to a WebSocket server using `Socket.IO`. Describe the effect of the code segment below.

```
var socket =
io('http://10.0.15.25:1234/deliverychat', {
    transports: ['websocket'] });
```

(4 marks)

Section B Q4. (Continued)

(c) By referring to the code segment below, construct the code so that the application can perform the following operations.

(i) When the application is connected to the server, the application displays a Toast message and send a boolean variable `connected` to the server where the value of `connected` is true. (6 marks)

(ii) Upon receiving the chat broadcast from the server, the React Native application first turns the data into a JavaScript object (refer to the **Table 4.1** for the keys in the object). The application then updates the state variable `liveChat` by appending the position and message. (6 marks)

Table 4.1: Updated Employee Information

Key	Value Type	Description
message	String	The message sent by the sender.
position	String	The position of the sender.

(iii) Upon receiving the error message from the server, the React Native application displays a Console message and navigate back to the previous screen. (6 marks)

(iv) The React Native application sends the `message` and `position` to the server where the value of `message` is from the state variable `message` and value of `position` is from the state variable `position`. (6 marks)

```
// Assume the necessary libraries were imported

var socket =
io('http://10.0.15.25:1234/deliverychat', {
    transports: ['websocket'],
});

const App = ({route, navigation}:any) => {

  const [position, setPosition] = useState('');
  const [liveChat, setLiveChat] = useState('');
  const [message, setMessage] = useState('');

  useEffect(()=>{

  // A new handler is registered
  /*
    Complete the code here for Q4 (c) (i)
  */
```

Section B Q4 (c) (iv) (Continued)

```
  // A new handler is registered
  /*
    Complete the code here for Q4 (c) (ii)
  */


  // A new handler is registered
  /*
    Complete the code here for Q4 (c) (iii)
  */

  },[]);

  return(
      <ScrollView>
              <TextInput
                   multiline = {true}
                   value = { liveChat }
              />
              <TextInput
                   multiline = {true}
                   value = { message }
                   onChangeText = { (input:
any) => {
                          setMessage(input);
                   } }
              />
              <Button
                   title = 'send'
                   onPress = {()=>{
             /*
             Complete the code here for Q4
(c) (iv)
             */
                   }}
              />
      </ScrollView>
  );
}
export default App;
```

[Total : 40 marks]